

Creating New Products and Services

Chapter 10 — Implementation Phase

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Learning Objectives

By the end of this chapter, you will be able to:

- **Understand** the transition from strategic selection to tactical implementation.
- **Analyze** the Stage-Gate™ model for managing development.
- **Contrast** Sequential vs. Simultaneous (Parallel) engineering.
- **Evaluate** the unique challenges of Service Innovation (NSD).
- **Apply** tools like Quality Function Deployment (QFD) and Rapid Prototyping.

The Implementation Challenge

We have selected the idea. Now we must build it.

New Product Development (NPD)

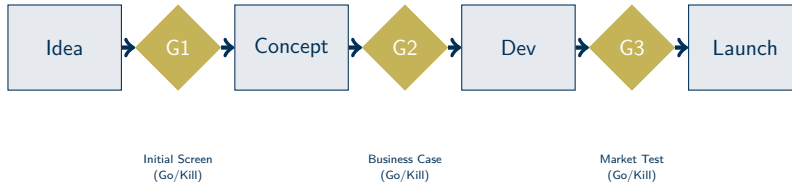
The process of bringing a new product or service to market. It is a high-risk activity where uncertainty is replaced by knowledge, but costs escalate rapidly.

Key Success Factors:

- A clear product definition (concept).
- A cross-functional team.
- A structured process (The "Stage-Gate").

The Stage-Gate System (Cooper)

A blueprint for managing the NPD process to control risk.

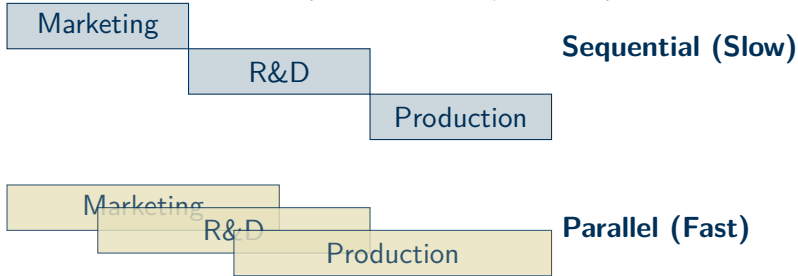


At each Gate, a decision is made: **Go, Kill, Hold, or Recycle.**

Accelerating the Process

Traditional (Sequential): "Over the wall" approach. Marketing finishes, then Engineering starts, then Manufacturing starts. Slow and error-prone.

Modern (Simultaneous/Parallel):



Developing Services (NSD)

Services are different from products.

Characteristics of Services

- **Intangibility:** You cannot "touch" a consultancy hour.
- **Simultaneity:** Production and consumption happen at the same time.
- **Heterogeneity:** Hard to standardize (depends on the person delivering it).

Implication: In NSD, we must design the *system* and the *customer experience*, not just the "item."

Tool: Quality Function Deployment (QFD)

How do we ensure the customer's voice is heard in engineering?

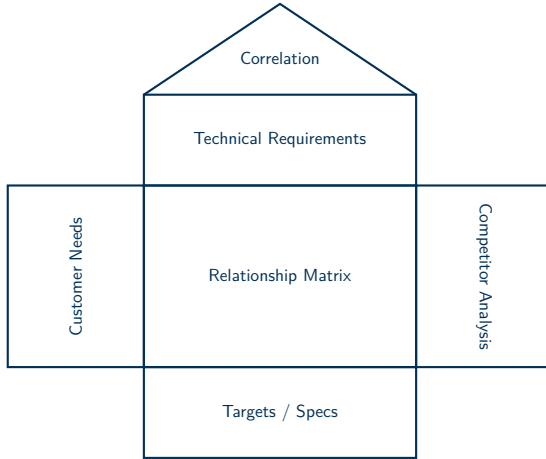
The House of Quality

A matrix tool that translates "Customer Attributes" (What they want) into "Engineering Characteristics" (How we build it).

Benefits:

- Reduces design changes later in the process.
- Ensures cross-functional alignment.
- Prioritizes technical features based on customer value.

The House of Quality Matrix



"Fail fast, fail cheap."

- **Purpose:** To learn, communicate, and integrate.
- **Evolution:**
 - *Low Fidelity:* Sketches, cardboard models, wireframes.
 - *High Fidelity:* 3D printing (Additive Manufacturing), Beta software versions.

The Cycle

Design → Build → Test → Analyze → Redesign

Lean Product Development

Inspired by Toyota, focused on minimizing waste.

Principles:

1. **Cadence:** Regular "heartbeat" of releases.
2. **Pull:** Don't start work until it's needed.
3. **Knowledge Reuse:** Capture lessons learned to avoid reinventing the wheel.
4. **Set-Based Design:** Keep multiple options open as long as possible before converging on the final solution.

Measuring NPD Performance

You cannot manage what you do not measure.

Efficiency Metrics:

- Time to Market (TTM).
- Development Cost vs. Budget.

Effectiveness Metrics:

- Customer Satisfaction.
- % of Revenue from products $\geq$ 3 years old.
- Break-even time.

Success Rate

Typically, only 1 in 7 concepts becomes a commercial success. The goal is to improve these odds.

Summary:

- Creating new products requires a disciplined **Implementation Process**.
- **Stage-Gate** systems manage risk by forcing decisions at critical points.
- **Cross-functional teams** working in **Parallel** speed up development.
- Tools like **QFD** and **Prototyping** ensure the final product meets customer needs.

Thank You

Questions?